Chapter 10

Quantum Geometry

In the course of about a decade, Einstein singlehandedly overthrew the centuries-old Newtonian framework and gave the world a radically new and demonstrably deeper understanding of gravity. It does not take much to get experts and nonexperts alike to gush over the sheer brilliance and monumental originality of Einstein's accomplishment in fashioning general relativity. Nevertheless, we should not lose sight of the favorable historical circumstances that strongly contributed to Einstein's success. Foremost among these are the nineteenth-century mathematical insights of Georg Bernhard Riemann that firmly established the geometrical apparatus for describing curved spaces of arbitrary dimension. In his famous 1854 inaugural lecture at the University of Göttingen, Riemann broke the chains of flat-space Euclidean thought and paved the way for a democratic mathematical treatment of geometry on all varieties of curved surfaces. It is Riemann's insights that provide the mathematics for quantitatively analyzing warped spaces such as those illustrated in Figures 3.4 and 3.6. Einstein's genius lay in recognizing that this body of mathematics was tailor-made for implementing his new view of the gravitational force. He boldly declared that the mathematics of Riemann's geometry aligns perfectly with the physics of gravity.

But now, almost a century after Einstein's tour-de-force, string theory gives us a quantum-mechanical description of gravity that, by necessity, modifies general relativity when the distances involved become as short as the Planck length. Since Riemannian geometry is the mathematical core of general relativity, this means that it too must be modified in order to reflect faithfully the new short-distance physics of string theory. Whereas general relativity asserts that the curved properties of the universe are described by Riemannian geometry, string theory asserts that this is true only if we examine the fabric of the universe on large enough scales. On scales as small as the Planck length a new kind of geometry must emerge, one that aligns with the new physics of string theory. This new geometrical framework is called quantum geometry.